Zero-Shot Cross-Carrier Transfer for Radar Micro-Doppler Human Activity Recognition

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***Abstract*—Radar micro-Doppler sensors recognize human activity, yet a recognizer trained on one radar fails on another that operates at a different carrier frequency, because the carrier stretches the Doppler signature by an unseen factor. Fielding each new sensor would then mean collecting and labeling data anew. This letter removes that cost in the carrier zero-shot setting, where the target sensor never appears in training, tuning, or model selection. The idea is to treat the carrier as a known physical quantity rather than an unknown domain: a Doppler-axis stretch (DAS) synthesizes how each recording would look at other carriers, and an adversarial carrier-residual (ACR) head trains the recognizer to ignore the carrier it still sees. Using only legacy 10 and 24 GHz archives, the method recognizes activity on an unseen 77 GHz radar at 0.832 macro-F1, against 0.302 for the strongest baseline, and holds 0.800 even when trained on carriers that never reach the test band, so an old low-band archive can field a new high-band sensor with no new data.**

***Index Terms*—Adversarial learning, domain generalization, human activity recognition, micro-Doppler radar, parameter-efficient fine-tuning, zero-shot transfer.**

I. Introduction

MICRO-DOPPLER spectrograms encode the limb kinematics of a moving person and support radar-based human activity recognition (HAR) in care, security, and automotive settings [1], [2]. Deployed sensors, however, span carrier bands: archived training data often comes from legacy 10 or 24 GHz radars, while new sensors operate at 77 GHz. For a scatterer with radial velocity v observed at carrier f_c, the Doppler shift is $f_d = 2vf_c/c$, so the entire Doppler axis of the spectrogram dilates in proportion to the carrier. A classifier trained on one band therefore meets, at test time, an input stretched by a factor it has never encountered. The shift is not a texture nuisance; it is a known parametric transformation.

Standard transfer pipelines do not survive this shift. On the benchmark used here (train 10+24 GHz, test 77 GHz, identical protocol for every method), eight public models spanning legacy and modern CNNs, hierarchical transformers, a radar-specific state-space model (RadMamba [3]), and a radar PEFT transformer (SelaFD [4]) reach 0.071 to 0.204 macro-F1, and a frozen DINOv3 control with our adapter and head reaches 0.302, despite source-validation scores of 0.91 to 0.98 for all. Discrete domain-adversarial training [5] does not close the gap either: the target carrier lies outside the training support, and a DANN control reaches 0.275 (Table II), no better than generic augmentation (0.302).

Prior radar work approaches the carrier gap from three directions. Cross-frequency adversarial training aligns the bands that are present in training [6]; physics-aware generative models synthesize additional training signatures [7]; and time- or Doppler-axis augmentation improves within-band generalization for gesture radar [8]; all three assume the test band, or its neighborhood, is represented in training. The zero-shot case, where the deployment carrier appears at no stage of training or selection, remains open; Section III shows strong models from these families collapse on it.

This letter treats the carrier as a continuous physical variable rather than a discrete domain label. Two components are added to a frozen DINOv3 ViT-L/16 [9] carrying a rank-2 LoRA adapter [10]. The first, Doppler-axis stretch (DAS), renders each source spectrogram at a virtual carrier f_v by rescaling its Doppler axis with the physical ratio f_v/f_s, applied along a curriculum that widens the virtual band as training progresses. The second, the adversarial carrier-residual (ACR) head, disentangles carrier from kinematics: a residual branch regresses the rendered carrier offset from a dedicated feature, while a gradient-reversal adversary regresses the continuous effective log-carrier from the recognition feature and thereby pushes that feature toward invariance along the carrier-scale direction. Evaluation is strictly inductive: the final exponential-moving-average (EMA) checkpoint at epoch 100 is used for every method, with no target data and no validation-based selection at any stage.

The contributions are as follows.

1) **Physics-driven carrier augmentation.** DAS lifts zero-shot 77 GHz macro-F1 from 0.302 (generic radar augmentation) to 0.767. The stretch operator alone reaches 0.704, while the same operator decoupled from the carrier ratio (over a narrower factor range) reaches only 0.518: the carrier-paired, wide-range stretch, not generic axis warping, carries the gain.
2) **Adversarial carrier-residual head.** ACR adds +6.55 pp over DAS alone (paired bootstrap 95% CI $[+4.69, +8.37]$), and its continuous log-carrier formulation beats an ordinary discrete domain adversary by +3.6 pp (0.832 vs. 0.796). The adversary is discarded at inference.
3) **Leakage-free zero-shot result.** The single model reaches 0.832 ± 0.034 macro-F1 on the full 418-clip 77 GHz test set, shrinking the generalization gap to +0.135 (all baselines: +0.67 or more). A three-seed

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posterior ensemble, reported separately as a deployment variant, reaches 0.856.

II. Proposed Method

A. Task, Notation, and Backbone

Training data consists of labeled micro-Doppler spectrograms $x \in \mathbb{R}^{224\times224\times3}$ (jet-rendered, per-image standardized) recorded at source carriers $f_s \in \{10, 24\}$ GHz with labels from seven activity classes. The model is evaluated zero-shot on an unseen 77 GHz sensor: no 77 GHz sample is used for training, hyperparameter tuning, or checkpoint selection. Fig. 1 summarizes the architecture.

A frozen DINOv3 ViT-L/16 [9] provides the representation; LoRA [10] factors of rank r=2 (α=8, dropout 0.10) on every block's qkv, projection, and MLP weights adapt the backbone. Together with the necks and heads, 2.4 M parameters train against the 0.3 B frozen weights; 1.9 M of them remain in the deployed recognition path, reported in Table I. Two parallel necks (LayerNorm, dropout 0.2, $1024 \rightarrow 512$ linear, GELU, LayerNorm) map the backbone feature to a recognition embedding z_{cls} and a carrier embedding z_{freq}, both 512-D. Classification uses an ArcFace cosine head [11] with logit adjustment [12]: the logit of class k is

$$\ell_k = s\,\hat{z}_{\mathrm{cls}}^{\top}\hat{W}_k - \tau\,\log\hat{p}(k), \tag{1}$$

where $\hat{z}_{\mathrm{cls}}$ and $\hat{W}_k$ are the ℓ_2-normalized embedding and class weight, s=24, τ=1, and $\hat{p}(k)$ is the empirical class prior; training subtracts an angular margin m=0.25 from the true-class cosine and applies label smoothing 0.05, giving the loss $\mathcal{L}_{\mathrm{cls}}$. A supervised contrastive term [13] on z_{cls} (weight 0.25, temperature 0.1) and a MIRO term [14] anchoring z_{cls} to a second, fully frozen copy of the backbone (weight 0.1) regularize the embedding.

B. DAS: Doppler-Axis Stretch

The micro-Doppler shift of a scatterer with radial velocity v at carrier f_c is

$$f_d = \frac{2vf_c}{c}, \tag{2}$$

where c is the speed of light, so $f_d \propto f_c$ for fixed kinematics. A clip recorded at f_s is rendered as if observed at a virtual carrier f_v by rescaling its Doppler (vertical) axis by

$$\rho = f_v/f_s, \tag{3}$$

resizing the height H to $\mathrm{round}(H\rho)$ and center-cropping back to H when $\rho > 1$, or center-padding with the corner-estimated background when $\rho < 1$. Eq. (3) presupposes a shared Doppler-frequency axis across bands (equal hertz per pixel, zero Doppler centered, common time grid) rather than velocity normalization, which would remove the cross-carrier gap; the uniform baseline collapse in Table I confirms the axes are carrier-scaled. Axis rescaling is an established radar augmentation [8]; DAS adds the carrier-matched parameterization of Eq. (3) and a schedule controlling how far the virtual band extrapolates. The virtual carrier is drawn log-uniformly from a band that widens on a three-stage curriculum: epochs 1–8 apply the stretch with probability 0.35 and $f_v \in [10, 24]$ GHz, epochs 9–24 with probability 0.70 and $f_v \in [10, 50]$ GHz, and epochs 25–100 with probability 1.0 and $f_v \in [12, 95]$ GHz. Two radar-specific augmentations are applied on top: a high-frequency Doppler texturizer (HFT; probability 0.65, uniform floor 0.05, Gaussian high-frequency noise 0.08 where the signal exceeds 0.15) and SpecAugment-style masks [15] (time masks to 10% width and Doppler masks to 12% height, each at probability 0.6). The endpoints span plausible deployment carriers rather than targeting a specific sensor, and no 77 GHz feedback entered any hyperparameter choice (Section III-C examines the schedule's interaction with the adversary).

C. ACR: Adversarial Carrier-Residual Head

ACR makes the carrier explicit and suppresses it in the recognition path. The residual branch regresses the rendered log-carrier offset $r = \log(f_v/f_s)$ from the carrier embedding,

$$\mathcal{L}_{\mathrm{freq}} = \mathrm{SmoothL1}\big(g(z_{\mathrm{freq}}),\, r\big), \tag{4}$$

where g is a linear head; the branch gives carrier information an explicit outlet. The adversarial branch acts on the recognition embedding through a gradient-reversal layer (GRL) [5] R_λ, which is the identity in the forward pass and multiplies gradients by $-\lambda$ in the backward pass:

$$\mathcal{L}_{\mathrm{grl}} = \mathrm{SmoothL1}\big(A(R_\lambda(z_{\mathrm{cls}})),\, \log f_s + r\big), \tag{5}$$

where A is a two-layer regressor ($512 \rightarrow 128 \rightarrow 1$, ReLU, dropout 0.2) and $\log f_s + r$ is the continuous effective log-carrier of the rendered input. The reversed gradient trains the encoder toward invariance of z_{cls} along the continuous Doppler-scale direction. The regression target is the decisive design choice: a conventional adversary classifies discrete domain bins, but the test carrier lies outside every training bin; Section III-C quantifies the difference. Adversarial regression over a continuous domain index echoes continuously indexed domain adaptation [16]; here no target samples exist, and the continuum is supplied by physics-rendered virtual carriers. The ramp $\lambda = 2/(1+e^{-10p})-1$ follows DANN, with p the fraction of completed training steps. The adversary is discarded at inference and excluded from the EMA.

The full training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + 0.25\,\mathcal{L}_{\mathrm{sc}} + 0.1\,\mathcal{L}_{\mathrm{miro}} + 0.05\,\mathcal{L}_{\mathrm{freq}} + 0.3\,\mathcal{L}_{\mathrm{grl}}, \tag{6}$$

where $\mathcal{L}_{\mathrm{sc}}$ and $\mathcal{L}_{\mathrm{miro}}$ are the supervised contrastive and MIRO terms of Section II-A.

D. Training and Selection Protocol

The proposed method, its ablations, and the generic baselines were trained with AdamW (learning rate 3×10^{-4}, weight decay 0.05), a cosine schedule with 3-epoch warmup, batch size 16, bf16 precision, class-balanced sampling, and an EMA of decay 0.999 from epoch 5, for 100 epochs; the radar-specific baselines kept their native optimizers (Section III-A). The reported checkpoint is always the final-epoch EMA weights for every method: no early stopping, no source-validation gate, no target peeking. The rule is motivated by saturation: source-validation F1 lies between 0.91 and 0.98 for every method in

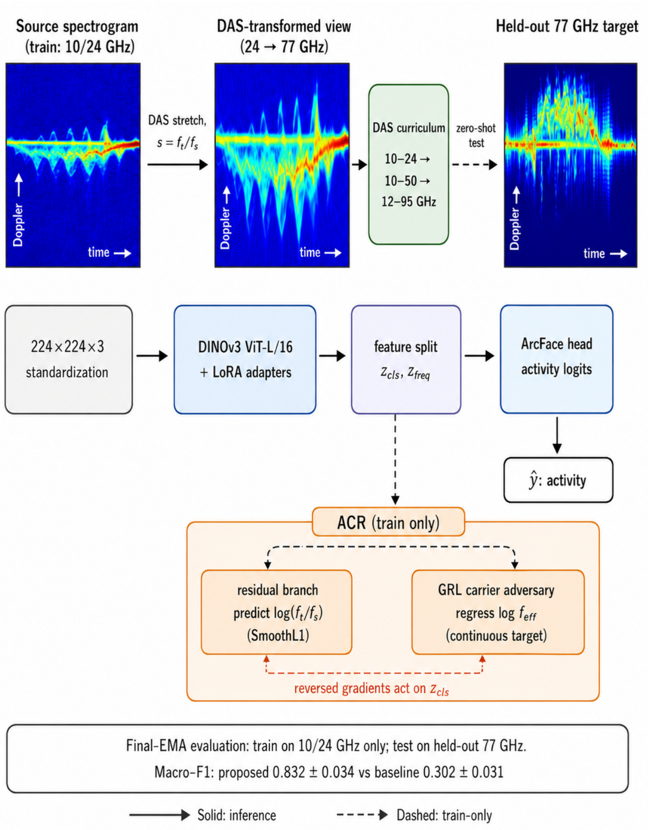

Fig. 1. Overview of the proposed cross-carrier method (DAS + ACR).

Table I while 77 GHz macro-F1 spans 0.071 to 0.832, so a source-validation gate carries no signal and a target gate would leak the test band. At test time the recognition head sees the raw spectrogram alone: no DAS, no test-time augmentation, no target-carrier value. Results are reported as mean ± std over three seeds (42, 1234, 31415).

E. Deployment Variant: Three-Seed Posterior Averaging

At deployment the softmax posteriors of the three seeded models are averaged before the argmax,

$$\bar{p}(k \mid x) = \tfrac{1}{3}\textstyle\sum_{s=1}^{3} \operatorname{softmax}\big(\ell^{(s)}(x)\big)_k, \qquad (7)$$

where $\ell^{(s)}$ denotes the logits of Eq. (1) from the s-th seed. Seed averaging is a standard variance-reduction step fixed a priori and uses no target information; Section III-C analyzes post hoc why it helps. The ensemble is reported separately throughout.

III. Experiments

A. Dataset and Protocol

Experiments use the public CI4R-MULTI3 corpus [6], [17] (co-located 10/24/77 GHz channels), merged at training with a 24 GHz supplement recorded at the National Research Council Canada: a Luswave PUP_EN24C_T2R4 FMCW radar with paired 15 dBi Tx/Rx horn antennas, mounted 1 m above the ground, captured six consenting volunteers performing the activities at 3 m on boresight, preserving the corpus's 1.6-second window and class taxonomy. Code, final-EMA checkpoints, and the complete training data are released at https://github.com/Awu-Lin/zero-shot-cross-carrier-radar-har. The three channels are an impulse-UWB Xethru X4 (center near 7.3 GHz, treated as the nominal 10 GHz band, matching the corpus labels), a 24 GHz Ancortek FMCW radar, and a 77 GHz mmWave FMCW radar; all obey the $f_d \propto f_c$ scaling DAS exploits. DAS computes its ratios from the nominal source values, so the 77 GHz figure enters training nowhere; the UWB channel's label mismatch is absorbed without special handling. The seven-class subset {Away, Bend, Kneel, Pick, side step (SStep), Sit, Towards} of the eleven-activity taxonomy is used: 644 training and 162 source-validation spectrograms at 10 and 24 GHz, and 418 test spectrograms at 77 GHz (the seven-class subset of the 653 held-out 77 GHz clips). The primary metric is macro-F1, the unweighted mean of per-class F1. Every 77 GHz test subject also appears in the source bands, so the carrier is the only unseen factor at test time; the NRC volunteers are source-only additions. Training ran on a single RTX 5090.

Every method follows the protocol of Section II-D. Generic image backbones were fully fine-tuned with the same optimizer recipe to avoid under-training them, with per-image standardization as the only preprocessing: no flipping (the time axis encodes the Away/Towards direction) and none of the proposed augmentations or domain-generalization losses. The gap to the control row in Table I, rather than to rows 1–8, is therefore the cleanest estimate of the contribution. Radar-specific baselines (RadMamba [3], SelaFD [4]) followed their native optimizer recipes under the same 100-epoch, final-EMA, no-selection protocol. The DINOv3-LoRA control shares our backbone, adapter, and head but removes DAS and ACR. The offline evaluator reproduces each run's training-history F1 exactly (max |diff| 0.0000).

TABLE I

Zero-shot 77 GHz results (3-seed mean ± std, final-EMA, 418-clip test set). Gap = source-val F1 − 77 GHz F1. Best single-model result in bold.

Method	Macro-F1	Acc.	Gap	Params (M)
VGG16-BN [18]	0.117 ± 0.009	0.212	+0.867	136.4
MobileNetV3-L [19]	0.146 ± 0.016	0.260	+0.828	4.5
EfficientNet-B0 [20]	0.089 ± 0.028	0.174	+0.824	4.3
ConvNeXt-T [21]	0.186 ± 0.024	0.282	+0.796	28.2
Swin-T [22]	0.071 ± 0.029	0.166	+0.908	27.9
ConvNeXtV2-T [23]	0.150 ± 0.016	0.210	+0.823	28.3
RadMamba [3]	0.204 ± 0.030	0.245	+0.721	0.1
SelaFD-ViT-B/16 [4]	0.107 ± 0.034	0.191	+0.805	14.5
DINOv3-L + LoRA [9]	0.302 ± 0.031	0.413	+0.668	1.9
Proposed (DAS+ACR)	$\mathbf{0.832 \pm 0.034}$	0.836	+0.135	1.9
Deployment variant (not head-to-head)				
+ 3-seed posterior ensemble	0.856	0.859	—	—

B. Main Comparison

Table I reports the head-to-head comparison. Every baseline collapses on the unseen carrier, the radar-specific ones included: macro-F1 spans 0.071 to 0.302 against saturated source validation, so the failure is purely a generalization gap, not a capacity problem. The proposed single model reaches

TABLE II
COMPONENT ABLATION ON ZERO-SHOT 77 GHz (3-SEED MEAN ± STD, FINAL-EMA, 418-CLIP TEST SET).

Variant	Acc.	Macro-F1
ERM (no augmentation)	0.271 ± 0.029	0.146 ± 0.034
+ radar aug. (HFT+SpecAug)	0.413 ± 0.025	0.302 ± 0.031
Physics-free axis jitter	0.585 ± 0.022	0.518 ± 0.030
DANN domain adversary [5]	0.374 ± 0.040	0.275 ± 0.024
Doppler stretch only	0.727 ± 0.075	0.704 ± 0.089
+ full DAS (stretch + radar aug.)	0.781 ± 0.070	0.767 ± 0.076
DAS + residual branch only	0.819 ± 0.021	0.814 ± 0.021
DAS + GRL adversary only	0.832 ± 0.016	0.826 ± 0.016
DAS + discrete-bin adversary	0.802 ± 0.032	0.796 ± 0.034
DAS + full ACR (proposed)	$\mathbf{0.836 \pm 0.032}$	$\mathbf{0.832 \pm 0.034}$

0.832 ± 0.034 macro-F1 (0.836 accuracy) and shrinks the gap to $+0.135$ from a source-validation F1 of 0.967. Against the DINOv3-LoRA control, which differs only by the absence of DAS and ACR, the gain is $+0.530$, a clean estimate of the cross-carrier contribution. Per-class F1 (3-seed means) spans 0.96 (Away) down to 0.72 (Sit), with Bend 0.79, Kneel and Pick 0.82, Towards 0.84, and SStep 0.88; the hardest classes remain the low-motion postures. The three-seed posterior ensemble reaches 0.856 macro-F1 (0.859 accuracy, bootstrap 95% CI $[0.821, 0.889]$) and is reported separately with 3× the single-model parameters and latency, since ensembling the baselines would be required for a fair head-to-head row.

The error structure of the single model is concentrated and anisotropic. Three confusions account for about 49% of all errors: Bend→Pick (23.2% of the Bend clips), Sit→Kneel (18.7%), and Towards→SStep (17.2%). The weak classes are high-precision but low-recall (e.g., Sit: precision 0.84, recall 0.64): the model under-predicts them, an anisotropic boundary rather than random degradation. A feature-level diagnosis explains the remaining headroom: a linear probe separating real 77 GHz clips from their DAS-rendered counterparts in the recognition features succeeds at 0.984, with the largest per-class separation on Sit (0.356) and Towards (0.232), the classes that dominate the confusions; the residue is consistent with a structural sensor gap, not a carrier failure.

C. Ablation and Significance

Table II isolates each component: the upper block varies the carrier-transfer mechanism, the lower block varies the carrier head on top of full DAS. In the augmentation body, generic radar augmentation lifts the ERM floor from 0.146 to only 0.302, whereas the Doppler stretch alone reaches 0.704, a $+55.8$ pp gain over ERM. The stretch must be paired with the carrier: the jitter control applies the identical rescale operator with $\rho \sim \mathrm{U}[0.8, 1.25]$ at probability 1.0, decoupled from any carrier ratio, and reaches only 0.518, trailing the stretch by 18.6 pp; carrier-blind warping over this narrower range is insufficient. The DANN control replaces the carrier mechanism with a binary source-band discriminator (10 versus 24 GHz) on the neck feature through a gradient-reversal layer (weight 0.1); with no target samples to align, it reaches 0.275. Stacking the radar augmentations on the stretch gives full DAS at 0.767, $+46.5$ pp over generic augmentation alone.

On top of DAS, each ACR half helps on its own, with the adversary dominant: the residual branch reaches 0.814, the GRL adversary 0.826, and the full head 0.832. The halves overlap: their gains are sub-additive ($+4.7$ and $+5.9$ against $+6.5$ pp jointly), so the two branches reach the same carrier information through complementary routes. Replacing the continuous log-carrier regression with a discrete 3-bin domain adversary drops performance to 0.796; treating the carrier as a continuous physical scale is worth $+3.6$ pp (95% CI $[+1.9, +5.3]$, paired stratified bootstrap, all replicates positive). The same-seed ACR-over-DAS deltas are $+12.0$, $+6.0$, and $+1.6$ pp, positive on all three seeds. A paired, class-stratified bootstrap over the 418 test clips ($B{=}10\,000$) confirms the gains: DAS+ACR over DAS is $+6.55$ pp (95% CI $[+4.69, +8.37]$, all replicates positive), and DAS over jitter is $+24.90$ pp (CI $[+21.47, +28.32]$). Scale is no substitute: unfreezing the full ViT-L instead of training LoRA collapses to 0.481 ± 0.075 even under a separate target-best protocol that favors the full fine-tune.

Two further controls address the schedule and the fusion rule. With ACR active, the curriculum is required for the full benefit: it reaches 0.832, a fixed narrow band ($[10, 30]$ GHz, never rendering the test carrier) gives 0.800, and a fixed full-width band ($[15, 140]$ GHz) collapses to 0.626; gradual widening keeps the adversarial game stable. The ensemble is rule-agnostic: majority voting, logit averaging, and posterior averaging yield nearly identical scores (0.851, 0.857, 0.856), and the paired bootstrap puts the posterior ensemble $+2.40$ pp over the single model (CI $[+0.89, +3.96]$, $P(\Delta > 0){=}0.999$). Per-class seed disagreement correlates with the per-class real-versus-rendered structural gap at $r = +0.79$ over the seven classes (Sit draws 49.1% disagreement; SStep none), so the seeds disagree exactly where the sensor gap is largest. Accordingly, the ensemble improves most on the hardest classes (Towards $+4.5$, Bend $+3.7$, Sit $+3.5$ pp) and cuts the three dominant confusions by 13.5%; the DAS-only configuration gains a similar $\approx +5$ pp, so the benefit is seed marginalization, not ACR.

IV. CONCLUSION

Cross-carrier collapse in radar micro-Doppler HAR is a parametric, physics-known shift, and this letter treats it as such: DAS renders virtual carriers from the $f_d \propto f_c$ law, and ACR adversarially suppresses the continuous carrier trace that survives rendering. Together they raise an unseen 77 GHz sensor from 0.302 to $0.832{\pm}0.034$ macro-F1 (ensemble 0.856) with 2.4 M trained parameters and no target data, and a fixed $[10, 30]$ GHz variant that never renders the test carrier still reaches 0.800, so the method extrapolates beyond its rendered band. Legacy low-band archives can therefore train a recognizer for a new high-band sensor without a single target-band clip. The remaining shortfall is confined to low-motion postures (Sit 0.72) and matches the residual sensor gap of Section III-B, not the carrier mechanism. All test subjects are seen during training, so the result isolates the carrier shift; joint subject-and-carrier shift and other target bands are the next steps.

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